NVIDIA Sybil — Reference Architecture (as the project documents it)

sybil runs ALONGSIDE the KDC and impersonates users by FORGING TGTs (reads keys from the KDC DB via kadm5) • MIT (LDAP backend) or FreeIPA / IdM

one Linux Kerberos realm — NOT Active Directory (AD is not in Sybil's supported environment; for the AD client adaptation see orthrus-architecture)

SINGLE KERBEROS REALM — MIT Kerberos (LDAP backend) or FreeIPA / RHEL IdM • all Linux

KDC TRUST CORE — co-located “alongside the KDC”

MIT KDC (LDAP backend) / FreeIPA IdM

- AS / TGS   (Kerberos, TCP 88)

KDB (LDAP backend) — sybild reads keys here

- principal keys incl. krbtgt ← read via kadm5 (forging)
- krbAllowedToDelegateTo { svc/... } + ok_as_delegate
  ← S4U ACLs for sybild's outbound delegation
- LDAP backend “required for S4U to work”

kadm5: read keys
(forge the user TGT)

sybild — privileged delegation daemon

- hosted ALONGSIDE the KDC — needs kadm5 / KDB access
- impersonation = FORGE a user TGT: reads the realm
  krbtgt key from the KDC DB (kadm5_get_principal_keys)
  and signs a hand-built ticket   [lib.rs → krbutil.c]
- also uses GSSAPI S4U for outbound service delegation
- /etc/sybil.toml ACLs • sybil keytab + kadm5 rights

store delegated creds

KCM — Kerberos Credential Manager

- “store delegated credentials” via the KCM protocol
- deployed alongside the KDC + sybild
- the credential store the job's creds come from

kinit -k (user)

sybil kinit user@DOMAIN

SPANK: forward + renew creds (--kerberos=yes)

HPC / SLURM

Login / submit node

- authorized principal: kinit -k
- sybil kinit user@DOMAIN.LAN   (CLI → sybild)
- authenticated delegation — not blanket unattended

submit

Slurm controller

- sbatch --kerberos=yes
- SPANK plugin hooks the job lifecycle

launch tasks

Compute nodes cn001..cnNNN

- SPANK plugin: forward + renew the user's creds
  across the job lifecycle (pulled from KCM)
- job authenticates as the user

use creds (GSSAPI)

Kerberized backends (same realm)

- NFS / Lustre / HDFS / …
- accept the delegated service tickets over GSSAPI

Kerberos / S4U (TCP 88) KDC-core internal HPC / Slurm internal Sybil credential delivery **Focal: sybild (amber)**

Source: github.com/NVIDIA/sybil